Factory Reallocation & Shipping Optimization Recommendation System

Executive Summary — Prepared for Government and Regulatory Stakeholders

Unified Mentor Data Science Internship - Project 2

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Purpose of This Report

This summary presents the findings of an analysis conducted to evaluate whether Nassau Candy Distributor's current factory-to-product assignment practices result in efficient shipping outcomes, and whether a data-driven reassignment approach could improve delivery performance without adversely affecting business viability. It is intended for readers who require a clear, accountable overview of the methodology, findings, and limitations, without requiring technical background in data science.

Scope and Basis of Analysis

This analysis is based on 10,194 historical order records, factory location data, and product-to-factory assignment records provided by Nassau Candy. During data preparation, it was identified that the original shipment date fields in the dataset did not reliably reflect real-world delivery durations. Rather than proceed using unreliable data, a defensible substitute measure was constructed using verified factory and customer geographic locations combined with standard shipping-method assumptions. This methodological decision, and its justification, is documented in full in the accompanying technical paper. Readers should note that all lead-time findings in this summary are derived from this constructed measure, not from directly observed shipment records.

Key Findings

- **Product concentration.** Approximately 97% of order volume is concentrated in a single product category (Chocolate), which is currently produced at only two of the five available factories. This concentration explains the majority of current shipping volume patterns.
- **Regional delivery disparity.** Customers in the Atlantic region experience the longest average delivery times, primarily attributable to geographic distance from currently assigned factories.
- **Potential for improvement.** Of 81 distinct shipping scenarios evaluated, 63 (77.8%) demonstrated a potential reduction in delivery time if reassigned to an alternative factory, with an average estimated reduction of 33.7% among those scenarios.
- **No adverse financial indication.** Statistical analysis found no meaningful relationship between shipping distance and order profitability (correlation coefficient of 0.007). This

indicates that reassignment for delivery-efficiency purposes is not expected to negatively affect financial outcomes, based on the data available.

- **Priority areas identified.** A subset of shipping routes was identified as both high-volume and consistently slower than average. These routes represent the highest-impact opportunity for improvement and are recommended as the starting point for further review.

Limitations and Caveats

This analysis is subject to several limitations that should be considered before any operational decisions are made:

- Delivery-time estimates are derived from a constructed measure, not directly observed shipment data (see Scope, above).
- Distance calculations reflect straight-line geographic distance and do not account for actual road or shipping routes, which are typically longer.
- Recommendations do not account for production capability, equipment, or the operational cost of reassigning a product to a new factory.
- Confidence in individual recommendations varies; recommendations supported by limited historical order volume are flagged separately in the accompanying dashboard and should be treated with additional caution.

Recommendation

It is recommended that Nassau Candy Distributor use this analysis as a starting point for operational review, prioritizing the high-volume, high-delay routes identified above. Any resulting reassignment decisions should incorporate production and operational feasibility review, which falls outside the scope of this analysis.